

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 2091

II

As specified in the official PDF

Source-grounded

Polymer Science

Study the official course objectives in the source syllabus.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 2091 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 2091.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma Programme
Course code	2091
Course title	Polymer Science
Semester	II
Course category	As specified in the official PDF
Periods per week	As specified in the official PDF
Periods per semester	As specified in the official PDF
Credits	As specified in the official PDF
Prerequisite	Topic Course code Course Title Semester Chemistry for Engineering Basic Chemistry-Types of bonding, Concept of Atomic weight 2003A I Practices Terminologies related to polymers : Monomer, Polymer,Oligomers, 1091 Introduction to Polymer Science I Functionality,Molecular weight of micromolecules and macromolecules Course Outcome CO Course Outcome Blooms Taxonomy Level CO1 Explain the chemistry of polymerization and various polymerization techniques. Understand CO2 Explain the concept of molecular weight and its determination. Understand CO3 Explain the concepts of glass transition temperature, crystallinity, and Stereo regularity. Understand CO4 Explain polymer degradation and polymer reactions. Understand CO-PO mapping COURSE ARTICULATION MATRIX PROGRAM OUTCOMES
Assessment	Use the official assessment section reproduced below

Modules

Topic entries

4 official module sections detected.

43 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES**Objectives, outcomes and mapping****Course objectives**

- Study the official course objectives in the source syllabus.

Course outcomes

CO	Official outcome	Study level
CO1	3 - - - - - 3	See official syllabus
CO2	3 - - - - - 3	See official syllabus
CO3	3 - - - - - 3	See official syllabus
CO4	3 - - - - - 3 Total 12 - - - - - 12 Correlation Level 3 - - - - - 3 Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-" Teaching and Assessment Scheme Examination Scheme Teaching Scheme/Week Self-learning Total Credits Theory Practical (SS)/Week Hours/Semester C Total L T P S CIE ESE Total CIE ESE Total Diploma Curriculum Revision 2026 2091 Page #1 4 0 0 0 6 60 4 40 60 100 0 0 0 100 L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination Syllabus - Major Topics Instructional Module Title Major Topics Hours L T • Polymerization ,Types of polymerization with examples- chain(addition) and step growth (condensation)polymerization. • Steps in Chain(addition) polymerization- initiation, propagation and termination, Types of initiators -free radicals, cations, or anions.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 - - - - - 3
CO2	CO2 3 - - - - - 3
CO3	CO3 3 - - - - - 3
CO4	CO4 3 - - - - - 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Polymerization reactions and techniques • Inhibitors, Shortstops and Chain transfer agents. 15 0	9	As specified
Module 2	15 0	10	As specified
Module 3	• Determination of Tg by dilatometry. 15 0	13	As specified
Module 4	• Polymer modification- Physical and Chemical methods 15 0	11	As specified

MODULE 1

Polymerization reactions and techniques • Inhibitors, Shortstops and Chain transfer agents. 15 0

This module is presented from the official syllabus scope for Polymer Science. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Polymerization reactions and techniques • Inhibitors, Shortstops and Chain transfer agents. 15 0
- Official duration: As specified
- Official topic entries captured: 9
- The full source excerpt is preserved below for coverage checking.

▼ Polymerization reactions and techniques • Inhibitors, Shortstops and Chain transfer agents. 15 0

Polymerization reactions and techniques • Inhibitors, Shortstops and Chain transfer agents. 15 0 is an official topic in Module 1 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Polymerization reactions and techniques • Inhibitors, Shortstops and Chain transfer agents. 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, polymerization reactions and techniques • inhibitors, shortstops and chain transfer agents. 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Polymerization reactions and techniques • Inhibitors, Shortstops and Chain transfer agents. 15 0 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പോളിമറൈസേഷൻ പ്രതികരണങ്ങളും രീതികളും • Inhibitors, Shortstops and Chain transfer agents. 15 0 പരീക്ഷയിൽ ഉപയോഗിക്കേണ്ട രീതികൾ definition/meaning പരിശോധിക്കുക. പരീക്ഷയിൽ ഉപയോഗിക്കേണ്ട, steps, application പരിശോധിക്കുക. Technical terms English-ൽ പരിശോധിക്കുക.

▼ • Mechanism of free radical polymerization with an example..

• Mechanism of free radical polymerization with an example.. is an official topic in Module 1 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Mechanism of free radical polymerization with an example.. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • mechanism of free radical polymerization with an example.. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What • Mechanism of free radical polymerization with an example.. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പ്രതിരോധം • Mechanism of free radical polymerization with an example.. പ്രതിരോധം definition/meaning പ്രതിരോധം. പ്രതിരോധം പ്രതിരോധം, steps, application പ്രതിരോധം പ്രതിരോധം. Technical terms English-പ്രതിരോധം പ്രതിരോധം.

▼ • Ionic polymerization, Co-ordination Polymerization.

• Ionic polymerization, Co-ordination Polymerization. is an official topic in Module 1 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Ionic polymerization, Co-ordination Polymerization. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • ionic polymerization, co-ordination polymerization. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Ionic polymerization, Co-ordination Polymerization. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഈ • Ionic polymerization, Co-ordination Polymerization. ഏതുതരം ഐോണിക് പ്രതിപ്രവർത്തനം definition/meaning എന്താണ്. ഏതുതരം പ്രതിപ്രവർത്തനം, steps, application എന്താണ്. Technical terms English-ഈ എന്താണ്.

▼ • Steps in Condensation polymerization.

• Steps in Condensation polymerization. is an official topic in Module 1 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Steps in Condensation polymerization. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • steps in condensation polymerization. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Steps in Condensation polymerization. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പദം • Steps in Condensation polymerization. പദം definition/meaning പദം. പദം പദം പദം, steps, application പദം പദം പദം. Technical terms English-പദം പദം പദം.

▼ • Polymerization techniques - Bulk, Solution , Suspension and

• Polymerization techniques - Bulk, Solution , Suspension and is an official topic in Module 1 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Polymerization techniques - Bulk, Solution , Suspension and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • polymerization techniques - bulk, solution , suspension and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Polymerization techniques - Bulk, Solution , Suspension and means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പൊളിമറൈസേഷൻ • Polymerization techniques - Bulk, Solution, Suspension and Emulsion. പൊളിമറൈസേഷൻ ന്റെ definition/meaning പൊളിമറൈസേഷൻ. പൊളിമറൈസേഷൻ ന്റെ steps, application പൊളിമറൈസേഷൻ പൊളിമറൈസേഷൻ. Technical terms English-ൽ പൊളിമറൈസേഷൻ പൊളിമറൈസേഷൻ.

▼ Emulsion polymerization.

Emulsion polymerization. is an official topic in Module 1 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Emulsion polymerization. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, emulsion polymerization. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Emulsion polymerization. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എം Emulsion polymerization. എംഎംഎംഎംഎംഎംഎംഎംഎം definition/meaning എംഎംഎംഎംഎംഎംഎം. എംഎംഎം എംഎംഎം എംഎംഎംഎം, steps, application എംഎംഎം എംഎംഎംഎംഎം എംഎംഎം. Technical terms English-എം എംഎംഎം എംഎംഎംഎംഎംഎം.

▼ • Distinguish molecular weight of micro molecules and

• Distinguish molecular weight of micro molecules and is an official topic in Module 1 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Distinguish molecular weight of micro molecules and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • distinguish molecular weight of micro molecules and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Distinguish molecular weight of micro molecules and means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പദം • Distinguish molecular weight of micro molecules and പദം definition/meaning പദം. പദം പദം പദം, steps, application പദം പദം പദം. Technical terms English-പദം പദം പദം.

▼ macromolecules.

macromolecules. is an official topic in Module 1 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify macromolecules. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, macromolecules. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What macromolecules. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പolymer macromolecules. പദപരിഭാഷണം പദപരിഭാഷണം definition/meaning പദപരിഭാഷണം. പദപരിഭാഷണം പദപരിഭാഷണം, steps, application പദപരിഭാഷണം പദപരിഭാഷണം പദപരിഭാഷണം. Technical terms English-പദപരിഭാഷണം പദപരിഭാഷണം.

▼ • Molecular weight of polymers - Average molecular weight

• Molecular weight of polymers - Average molecular weight is an official topic in Module 1 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Molecular weight of polymers - Average molecular weight using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • molecular weight of polymers - average molecular weight should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Molecular weight of polymers - Average molecular weight means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പദം • Molecular weight of polymers - Average molecular weight പദത്തിന്റെ definition/meaning പദത്തിന്റെ definition. പദത്തിന്റെ പദത്തിന്റെ പദത്തിന്റെ, steps, application പദത്തിന്റെ പദത്തിന്റെ പദത്തിന്റെ. Technical terms English-പദം പദത്തിന്റെ പദത്തിന്റെ.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2**15 0**

This module is presented from the official syllabus scope for Polymer Science. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: 15 0
- Official duration: As specified
- Official topic entries captured: 10
- The full source excerpt is preserved below for coverage checking.

▼ Weight • Different molecular weights of polymers - M_n , M_w , M_v and M_z

Weight • Different molecular weights of polymers - M_n , M_w , M_v and M_z is an official topic in Module 2 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Weight • Different molecular weights of polymers - M_n , M_w , M_v and M_z using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, weight • different molecular weights of polymers - m_n , m_w , m_v and m_z should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Weight • Different molecular weights of polymers - M_n , M_w , M_v and M_z means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- $\square\square$ Weight • Different molecular weights of polymers - M_n , M_w , M_v and M_z $\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square$, steps, application $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square\square$.

▼ • Derive equations for M_n and M_w .

• Derive equations for M_n and M_w . is an official topic in Module 2 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Derive equations for M_n and M_w . using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • derive equations for m_n and m_w . should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Derive equations for M_n and M_w . means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പോളിമർ • Derive equations for M_n and M_w .
 പൊളിമർ തരംഗത്തിന്റെ definition/meaning പൊളിമർ തരംഗത്തിന്റെ. പൊളിമർ തരംഗത്തിന്റെ
 പൊളിമർ തരംഗത്തിന്റെ, steps, application പൊളിമർ തരംഗത്തിന്റെ പൊളിമർ തരംഗത്തിന്റെ. Technical terms
 English-ൽ പൊളിമർ തരംഗത്തിന്റെ പൊളിമർ തരംഗത്തിന്റെ.

▼ • Polydispersity and molecular weight distribution.

• Polydispersity and molecular weight distribution. is an official topic in Module 2 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Polydispersity and molecular weight distribution. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • polydispersity and molecular weight distribution. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Polydispersity and molecular weight distribution. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പോളിമർ • Polydispersity and molecular weight distribution. പോളിമർ പോളിമർ definition/meaning പോളിമർ പോളിമർ. പോളിമർ പോളിമർ പോളിമർ, steps, application പോളിമർ പോളിമർ പോളിമർ. Technical terms English-പോളിമർ പോളിമർ പോളിമർ.

▼ • **Methods of determination of molecular weight of polymers.**

• Methods of determination of molecular weight of polymers. is an official topic in Module 2 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Methods of determination of molecular weight of polymers. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • methods of determination of molecular weight of polymers. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What • Methods of determination of molecular weight of polymers. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പോളിമർ • Methods of determination of molecular weight of polymers. പൊളിമർ തന്മാത്രാ ഭാരം definition/meaning പൊളിമർ തന്മാത്രാ ഭാരം. പൊളിമർ തന്മാത്രാ ഭാരം, steps, application പൊളിമർ തന്മാത്രാ ഭാരം. Technical terms English-പോളിമർ തന്മാത്രാ ഭാരം.

▼ • Amorphous, Crystalline and semi-crystalline polymers with

• Amorphous, Crystalline and semi-crystalline polymers with is an official topic in Module 2 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Amorphous, Crystalline and semi-crystalline polymers with using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • amorphous, crystalline and semi-crystalline polymers with should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Amorphous, Crystalline and semi-crystalline polymers with means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പദം • Amorphous, Crystalline and semi-crystalline polymers with പദത്തിന്റെ definition/meaning പദം. പദം പദം, steps, application പദം പദം. Technical terms English-പദം പദം.

▼ examples.

examples. is an official topic in Module 2 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify examples. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, examples. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What examples. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഉദാഹരണങ്ങൾ. നിർവ്വചനം/അർത്ഥം ഉദാഹരണങ്ങൾ. ഘട്ടങ്ങൾ, steps, application ഉദാഹരണങ്ങൾ ഉദാഹരണം. Technical terms English-ൽ ഉദാഹരണങ്ങൾ ഉദാഹരണം.

▼ • Glass Transition temperature (Tg), Melting temperature (Tm),

• Glass Transition temperature (Tg), Melting temperature (Tm), is an official topic in Module 2 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Glass Transition temperature (Tg), Melting temperature (Tm), using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • glass transition temperature (tg), melting temperature (tm), should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Glass Transition temperature (Tg), Melting temperature (Tm), means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പ്ലാസ്റ്റിക് • Glass Transition temperature (T_g), Melting temperature (T_m), പ്ലാസ്റ്റിക് വികാസം definition/meaning പ്ലാസ്റ്റിക് വികാസം. പ്ലാസ്റ്റിക് വികാസം പ്ലാസ്റ്റിക്, steps, application പ്ലാസ്റ്റിക് വികാസം പ്ലാസ്റ്റിക്. Technical terms English-പ്ലാസ്റ്റിക് വികാസം പ്ലാസ്റ്റിക് വികാസം.

▼ Flow temperature(Tf), Decomposition temperature(Td).

Flow temperature(Tf), Decomposition temperature(Td). is an official topic in Module 2 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Flow temperature(Tf), Decomposition temperature(Td). using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, flow temperature(tf), decomposition temperature(td). should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Flow temperature(Tf), Decomposition temperature(Td). means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഫ്ലോ താപനില (Tf), Decomposition temperature (Td). ഫ്ലോ താപനില definition/meaning ഫ്ലോ താപനില. ഫ്ലോ താപനില ഫ്ലോ താപനില, steps, application ഫ്ലോ താപനില ഫ്ലോ താപനില. Technical terms English-ഫ്ലോ താപനില ഫ്ലോ താപനില.

▼ • Factors affecting Tg

• Factors affecting Tg is an official topic in Module 2 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Factors affecting Tg using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • factors affecting tg should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Factors affecting Tg means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഘടകങ്ങൾ • Factors affecting Tg നിർണ്ണയിക്കുന്ന ഘടകങ്ങൾ definition/meaning നിർവ്വചന/അർത്ഥം. ഘടകങ്ങൾ ഘടനാപരമായ ഘടകങ്ങൾ, steps, application പ്രയോഗം നിർവ്വചനാപരമായ ഘടകങ്ങൾ. Technical terms English-ൽ നിർവ്വചനം നിർവ്വചനാപരമായ ഘടകങ്ങൾ.

▼ Glass Transition

Glass Transition is an official topic in Module 2 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Glass Transition using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, glass transition should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Glass Transition means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഗ്ലാസ് ട്രാൻസിഷൻ ടെമ്പറേച്ചർ നോട്ട്സ് definition/meaning നോട്ട്സ്. ടെമ്പറേച്ചർ നോട്ട്സ്, steps, application നോട്ട്സ്. Technical terms English-ൽ നോട്ട്സ്.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3**• Determination of Tg by dilatometry. 15 0**

This module is presented from the official syllabus scope for Polymer Science. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: • Determination of Tg by dilatometry. 15 0
- Official duration: As specified
- Official topic entries captured: 13
- The full source excerpt is preserved below for coverage checking.

▼ • Determination of Tg by dilatometry. 15 0

• Determination of Tg by dilatometry. 15 0 is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Determination of Tg by dilatometry. 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • determination of tg by dilatometry. 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Determination of Tg by dilatometry. 15 0 means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പദം • Determination of Tg by dilatometry. 15 0 പദം/പദം definition/meaning പദം/പദം. പദം/പദം പദം/പദം പദം/പദം, steps, application പദം/പദം പദം/പദം പദം/പദം. Technical terms English-പദം പദം/പദം പദം/പദം.

▼ Temperature,Crystallinity

Temperature,Crystallinity is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Temperature,Crystallinity using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, temperature,crystallinity should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Temperature,Crystallinity means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- $\square\square$ Temperature, Crystallinity $\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square$. $\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square\square$, steps, application $\square\square\square\square$ $\square\square\square\square\square\square\square$ $\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square$.

▼ • Comparison of Elastomers, plastics and fibers in terms of Tg and

• Comparison of Elastomers, plastics and fibers in terms of Tg and is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Comparison of Elastomers, plastics and fibers in terms of Tg and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • comparison of elastomers, plastics and fibers in terms of tg and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What • Comparison of Elastomers, plastics and fibers in terms of Tg and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പ്രതി • Comparison of Elastomers, plastics and fibers in terms of Tg and പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-ൽ പദപരിചയം പദപരിചയം.

▼ crystallinity.

crystallinity. is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify crystallinity. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, crystallinity. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What crystallinity. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ക്രystallinity. ക്രystallinity ന്റെ definition/meaning എഴുതുക. ക്രystallinity ന്റെ application എഴുതുക. steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

▼ • Stereo regular polymers- Optical isomerism and Geometric

• Stereo regular polymers- Optical isomerism and Geometric is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Stereo regular polymers- Optical isomerism and Geometric using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • stereo regular polymers- optical isomerism and geometric should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What • Stereo regular polymers- Optical isomerism and Geometric means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പദം • Stereo regular polymers- Optical isomerism and Geometric പദങ്ങൾ definition/meaning പദങ്ങൾ. പദങ്ങൾ പദങ്ങൾ പദങ്ങൾ, steps, application പദങ്ങൾ പദങ്ങൾ പദങ്ങൾ. Technical terms English-പദങ്ങൾ പദങ്ങൾ പദങ്ങൾ.

▼ isomerism

isomerism is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify isomerism using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, isomerism should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What isomerism means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഇംഗ്ലീഷ് isomerism എന്നതിന്റെ definition/meaning എഴുതുക. ഇതിന്റെ ഉദാഹരണങ്ങൾ, steps, application എന്നിവയെക്കുറിച്ചും എഴുതുക. Technical terms English-ൽ എഴുതുക.

▼ • Molecular structure of IR, BR and CR

• Molecular structure of IR, BR and CR is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Molecular structure of IR, BR and CR using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • molecular structure of ir, br and cr should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Molecular structure of IR, BR and CR means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പദം • Molecular structure of IR, BR and CR
 പദത്തിന്റെ definition/meaning പദം. പദം പദം
 പദം, steps, application പദം പദം. Technical terms
 English-പദം പദം.

▼ • Polymer reactions - Differentiate reactions of polymers with

• Polymer reactions - Differentiate reactions of polymers with is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Polymer reactions - Differentiate reactions of polymers with using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • polymer reactions - differentiate reactions of polymers with should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Polymer reactions - Differentiate reactions of polymers with means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പോളിമർ • Polymer reactions - Differentiate reactions of polymers with പൊളിമർ രാസപ്രതികരണങ്ങൾ definition/meaning പൊളിമർ രാസപ്രതികരണങ്ങൾ. പൊളിമർ രാസപ്രതികരണങ്ങൾ, steps, application പൊളിമർ രാസപ്രതികരണങ്ങൾ. Technical terms English-പോളിമർ രാസപ്രതികരണങ്ങൾ.

▼ simple molecules.

simple molecules. is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify simple molecules. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, simple molecules. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What simple molecules. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എളുപ്പമായ തന്മാത്രകൾ. തന്മാത്രയുടെ നിർവ്വചനം/അർത്ഥം എഴുതുക. തന്മാത്രയുടെ ഉപയോഗം, steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

▼ • Effect of functional groups in polymers - Hydroxyl - Carboxylic -

• Effect of functional groups in polymers - Hydroxyl - Carboxylic - is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Effect of functional groups in polymers - Hydroxyl - Carboxylic - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • effect of functional groups in polymers - hydroxyl - carboxylic - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What • Effect of functional groups in polymers - Hydroxyl - Carboxylic - means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പ്രതിരോധം • Effect of functional groups in polymers - Hydroxyl - Carboxylic - പ്രതിരോധം പ്രതിരോധം definition/meaning പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം, steps, application പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം. Technical terms English-ൽ പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം.

▼ Aldehydic groups, Addition and Substitution reactions, Cross

Aldehydic groups, Addition and Substitution reactions, Cross is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Aldehydic groups, Addition and Substitution reactions, Cross using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, aldehydic groups, addition and substitution reactions, cross should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Aldehydic groups, Addition and Substitution reactions, Cross means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓൾഡൈഡിക് ഗ്രൂപ്പ്, Addition and Substitution reactions, Cross ഓൾഡൈഡിക് ഗ്രൂപ്പ് definition/meaning ഓൾഡൈഡിക് ഗ്രൂപ്പ്. ഓൾഡൈഡിക് ഗ്രൂപ്പ് ഓൾഡൈഡിക്, steps, application ഓൾഡൈഡിക് ഓൾഡൈഡിക് ഓൾഡൈഡിക്. Technical terms English-ഓൾഡൈഡിക് ഓൾഡൈഡിക്.

▼ linking reactions.

linking reactions. is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify linking reactions. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, linking reactions. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What linking reactions. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- π linking reactions. π π definition/meaning π . π π π , steps, application π π . Technical terms English- π π π .

▼ Polymer reactions, modifications of

Polymer reactions, modifications of is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Polymer reactions, modifications of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, polymer reactions, modifications of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Polymer reactions, modifications of means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പോളിമർ പ്രതികരണങ്ങൾ, മാറ്റങ്ങൾ of പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-ൽ പദപരിചയം പദപരിചയം.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

• Polymer modification- Physical and Chemical methods 15 0

This module is presented from the official syllabus scope for Polymer Science. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: • Polymer modification- Physical and Chemical methods 15 0
- Official duration: As specified
- Official topic entries captured: 11
- The full source excerpt is preserved below for coverage checking.

▼ • Polymer modification- Physical and Chemical methods 15 0

• Polymer modification- Physical and Chemical methods 15 0 is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Polymer modification- Physical and Chemical methods 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • polymer modification- physical and chemical methods 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Polymer modification- Physical and Chemical methods 15 0 means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പോളിമർ • Polymer modification- Physical and Chemical methods 15 0 പദപരിഭാഷ, definition/meaning പദപരിഭാഷ. പദപരിഭാഷ പദപരിഭാഷ, steps, application പദപരിഭാഷ പദപരിഭാഷ. Technical terms English-പദപരിഭാഷ പദപരിഭാഷ.

▼ polymer and polymer degradation.

polymer and polymer degradation. is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify polymer and polymer degradation. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, polymer and polymer degradation. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What polymer and polymer degradation. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പോളിമർ and polymer degradation. പൊതുവെ പോളിമർ definition/meaning പൊതുവെ. പോളിമർ പോളിമർ പോളിമർ, steps, application പോളിമർ പോളിമർ പോളിമർ. Technical terms English-ൽ പോളിമർ പോളിമർ പോളിമർ.

▼ • Polymer Degradation- Chain and random degradation- Factors

• Polymer Degradation- Chain and random degradation- Factors is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Polymer Degradation- Chain and random degradation- Factors using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • polymer degradation- chain and random degradation- factors should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Polymer Degradation- Chain and random degradation- Factors means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പോളിമർ • Polymer Degradation- Chain and random degradation- Factors പാലിമർ വിഘടനം definition/meaning പാലിമർ വിഘടനം. പാലിമർ വിഘടനം പാലിമർ, steps, application പാലിമർ വിഘടനം പാലിമർ. Technical terms English-പാലിമർ വിഘടനം.

▼ affecting polymer degradation

affecting polymer degradation is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify affecting polymer degradation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, affecting polymer degradation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What affecting polymer degradation means in the official course context
Study lens	Principle → components or stages → result → application

Definition, labelled explanation, comparison, procedure or example as appropriate

000000000000 00000 definition/meaning 000000000000. 000000 000000
 00000000, steps, application 000000 0000000000 000000. Technical terms
 English-0 00000 000000000000.

▼ • Types of degradation

• Types of degradation is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Types of degradation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • types of degradation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What • Types of degradation means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പദം • Types of degradation പദപരിവർത്തനം പദപരിവർത്തനം definition/meaning പദപരിവർത്തനം. പദപരിവർത്തനം പദപരിവർത്തനം, steps, application പദപരിവർത്തനം പദപരിവർത്തനം പദപരിവർത്തനം. Technical terms English-പദ പദപരിവർത്തനം പദപരിവർത്തനം.

▼ • Prevention of degradation- Antioxidant, Thermal stabilizers, U.V.

• Prevention of degradation- Antioxidant, Thermal stabilizers, U.V. is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify • Prevention of degradation- Antioxidant, Thermal stabilizers, U.V. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, • prevention of degradation- antioxidant, thermal stabilizers, u.v. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What • Prevention of degradation- Antioxidant, Thermal stabilizers, U.V. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പ്രതിരോധം • Prevention of degradation- Antioxidant, Thermal stabilizers, U.V. പ്രതിരോധത്തിന്റെ നിർവ്വചനം/അർത്ഥം പ്രതിരോധത്തിന്റെ തത്വം. പ്രതിരോധത്തിന്റെ ഘടകങ്ങൾ, steps, application പ്രതിരോധത്തിന്റെ ഉദാഹരണം. Technical terms English-ൽ പ്രതിരോധം പ്രതിരോധത്തിന്റെ.

▼ Stabilizers.

Stabilizers. is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Stabilizers. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, stabilizers. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Stabilizers. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-സ്റ്റാബിലൈസറുകൾ. സ്റ്റാബിലൈസറുകളുടെ നിർവ്വചനം/അർത്ഥം, സ്റ്റാബിലൈസറുകളുടെ പ്രയോഗം, steps, application എന്നിവയെക്കുറിച്ചുള്ള വിവരങ്ങൾ. Technical terms English-ൽ പരിഭാഷിക്കുക.

▼ Detailed Syllabus

Detailed Syllabus is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Detailed Syllabus using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, detailed syllabus should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Detailed Syllabus means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്ന Detailed Syllabus പരീക്ഷാസൂചികയിൽ നിന്ന് definition/meaning പരീക്ഷാസൂചികയിൽ. പരീക്ഷാ സൂചികയിൽ പരീക്ഷാസൂചിക, steps, application പരീക്ഷാ സൂചികയിൽ പരീക്ഷാസൂചിക. Technical terms English-ൽ പരീക്ഷാ സൂചികയിൽ പരീക്ഷാസൂചിക.

▼ Mode of

Mode of is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mode of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mode of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mode of means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നിവിടെ Mode of ഏകീകരണം എന്നതിന് definition/meaning എഴുതണം. ഏകീകരണം എങ്ങനെ നടത്തണം, steps, application എന്നിവയെക്കുറിച്ച് എഴുതണം. Technical terms English-ൽ എഴുതണം.

▼ Mapped Instructional

Mapped Instructional is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mapped Instructional using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mapped instructional should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mapped Instructional means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നിവിടെ Mapped Instructional ഉള്ളതാണ്. definition/meaning എഴുതുക. ഉദാഹരണം, steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

▼ Subtopic Student Learning Outcome delivery Taxonomy Level

Subtopic Student Learning Outcome delivery Taxonomy Level is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Subtopic Student Learning Outcome delivery Taxonomy Level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, subtopic student learning outcome delivery taxonomy level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Subtopic Student Learning Outcome delivery Taxonomy Level means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-Subtopic Student Learning Outcome delivery Taxonomy Level definition/meaning. Steps, application. Technical terms English-.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

- for Self-Learning
- Week Self-Learning description Hours Module I
- Create a table listing monomers, mechanism, by-products, and examples for addition and step growth polymerization. 6
- Watch a polymerization reaction video or simulation and note the differences between chain and step-growth processes. 6
- Create a table comparing bulk, solution, suspension, emulsion polymerization for monomer type, advantages,
- disadvantages, and examples. 6
- Use online simulations to observe polymerization kinetics in different techniques and note differences.
- Use websites like PhET, ChemCollective, or LearnChemE to simulate polymerization reactions. 6 Module II
- Prepare a one-page summary on importance of molecular weight in polymer processing. 6
- Interpret molecular weight distribution curves and identify M_n , M_w , and M_z on the graph. 6
- Explains one molecular weight determination method (osmometry, viscometry, light scattering). 6
- Find M_n and M_w of common polymers like Polyethylene (PE) ,Polystyrene (PS) etc. and see how PDI affects mechanical
- properties. Module III
- Make a table of common polymers, their T_g , T_m , and crystallinity. 6
- Collect different plastic, rubber, and fiber samples from home. Classify them as amorphous or crystalline. 6
- Use beads or colored clay to make polymer chains showing isotactic, syndiotactic, and atactic arrangements. 6
- Comparison Chart: Elastomers, Plastics, and Fibers - Prepare a table comparing: T_g range, Degree of crystallinity, Mechanical
- behavior. Module IV

- Find a commercial polymer blend or alloy and explain why it was modified. 6
- Take two separate polymers (e.g., rubber and plastic) and imagine combining them as an IPN. Predict the changes in their

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

– Theory						
CA3	CA4	CA5	CA1	CA6	CA2	ESE
Mode	Attendance		Self-learning	Self-learning		
Self-learning	Self-learning		Series	Series		
Written						
activities	activities		activities		activities	
Exam		Exam	Exam			
(Module 3)	(Module 4)	(Module 1)	(Module 2)			
(theory)		(2 modules)	(2 modules)			
Duration						
2 hours	2 hours	2.5 hours				
Exam mark			15		15	
15	15	50		50		60
Converted to						
20	20					
Marks		5				
15				20		
60						
Total						
40						
60						
Tentative	End of					
End of						

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► Define or explain Polymerization reactions and techniques • Inhibitors, Shortstops and Chain transfer agents. 15 0. (3 marks)

► Define or explain • Mechanism of free radical polymerization with an example... (3 marks)

► Define or explain • Ionic polymerization, Co-ordination Polymerization.. (3 marks)

► Define or explain • Steps in Condensation polymerization.. (3 marks)

Module 2

► Define or explain Weight • Different molecular weights of polymers - M_n , M_w , M_v and M_z . (3 marks)

► Define or explain • Derive equations for M_n and M_w .. (3 marks)

► Define or explain • Polydispersity and molecular weight distribution.. (3 marks)

► Define or explain • Methods of determination of molecular weight of polymers.. (3 marks)

Module 3

► Define or explain • Determination of Tg by dilatometry. 15 0. (3 marks)

► Define or explain Temperature, Crystallinity. (3 marks)

► Define or explain • Comparison of Elastomers, plastics and fibers in terms of Tg and. (3 marks)

► Define or explain crystallinity.. (3 marks)

Module 4

► Define or explain • Polymer modification- Physical and Chemical methods 15 0. (3 marks)

► Define or explain polymer and polymer degradation.. (3 marks)

► Define or explain • Polymer Degradation- Chain and random degradation- Factors. (3 marks)

► Define or explain affecting polymer degradation. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Polymerization reactions and techniques • Inhibitors, Shortstops and Chain transfer agents. 15 0.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Weight • Different molecular weights of polymers - Mn, Mw, Mv and Mz.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **• Determination of Tg by dilatometry. 15 0.**

3-mark explanation: Explain one principle, process or comparison from

Module 4

1-mark definition: State the meaning of **• Polymer modification- Physical and Chemical methods 15 0.**

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- URL Modules Covered
- <https://learncheme.com/screencasts/materials-science/> I
- [https://www.open.edu/openlearn/science-maths-technology/chemistry/introduction-polymers I & II](https://www.open.edu/openlearn/science-maths-technology/chemistry/introduction-polymers-I-&II)
- <https://pslc.ws/macrog/maindir.htm> All
- <https://ocw.mit.edu/courses/10-569-synthesis-of-polymers-fall-2006/> All

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTR syllabus PDF for Course 2091. Verify institutional updates against the current official curriculum.